

PIPEWORK & FITTINGS

surrounding structure, without the need for flexible connections. Where this is not possible, and in any case where the pipes are connected to equipment such as chillers, pumps and air handling units, approved type flexible pipe connections with all necessary guides of standard proprietary make shall be used.

4.25 EXPANSION JOINTS

- a. Wherever necessary, due allowance shall be made for the expansion and contraction of pipework by change in direction of pipework, or expansion loops, whether shown on the Drawings or not.
- b. Expansion joints shall be of the flanged type manufactured of high grade neoprene suitable for the temperature and pressures involved.
- c. The type and location of the expansion joints shall be submitted to the Architect for review prior to installation. Where pipes pass through building expansion joints, bellow type expansion joints shall be provided.

4.26 UNDERGROUND PIPE PROTECTION

Noted

- a. All underground pipework of copper or steel construction where located below ground or within corrosive areas shall be adequately protected with a protective double wrapping with 2 layers of petrolatum tape.
- b. The first application shall be installed with a minimum 50% overlap when covered in spiral formation over the pipe and the second application with a minimum 25% overlap when covered in a reverse spiral formation over the previous layer. Tape widths shall be as recommended by the manufacturer for the size of pipe to be covered but shall not be less than 75mm wide for pipework up to 50mm diameter.

4.27 PIPEWORK INSULATION

- a. Insulation for the various services shall be according to the attached Table 3.
- b. The following types of insulation shall be employed :

| <u>Class of Insulation</u> | <u>Description</u> |
|----------------------------|--------------------|
|----------------------------|--------------------|



## MUMBAI HIGH END RESIDENTIAL BUILDING | MUMBAI INDIA

### SECTION 4 ACMV INSTALLATION

### GENERAL SPECIFICATION

#### PIPEWORK & FITTINGS

A Sectional moulded closed cell polyurethane foam of the self extinguishing type, having a thermal conductivity after 6 months ageing of not greater than  $0.02^{\circ}\text{W/m}^{\circ}\text{C}$  at a mean temperature of  $25^{\circ}\text{C}$ . Minimum density and compressive strength shall be  $25 \text{ kg/m}^3$  and  $150 \text{ kPA}$ . Water permeability shall be less than  $1.8 \times 10^{-5}$  metric perms.

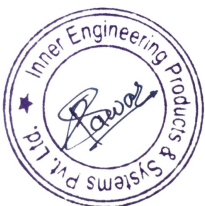
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B Flexible closed cell chemically blown PVC nitrite rubber sponge type having a thermal conductivity of not greater than  $0.037 \text{ W/m}^{\circ}\text{C}$  at a mean temperature of  $24^{\circ}\text{C}$  and shall have a water vapour permeability factor of  $M \geq 3,500$ . Flame spread fire performance shall be Class I with self extinguishing fire properties.

c. The following minimum insulation thickness shall be employed to minimise heat loss and prevent condensation. Heat loss calculation and condensation calculation based on fluid temperature, surrounding with temperature and humidity shall be submitted to the Architect for approval to determine the required insulation thickness.

| Nominal<br>Pipe Size<br>mm | Chilled<br>Water<br>Pipe | Condensate<br>Water Pipe | Refrigerant<br>Pipe | Condenser<br>Water pipe<br>exposed in air-<br>conditioned<br>area/plenum |
|----------------------------|--------------------------|--------------------------|---------------------|--------------------------------------------------------------------------|
| Up to 25                   | 25                       | 20                       | 25                  | 20                                                                       |
| 32 to 40                   | 25                       | 20                       | 40                  | 20                                                                       |
| 50 to 200                  | 40                       | 25                       | -                   | 20                                                                       |
| 200<br>above               | 50                       | -                        | -                   | 20                                                                       |
| Exposed<br>to<br>ambient   | 75                       | 25                       | 40                  | -                                                                        |

Noted



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### GENERAL SPECIFICATION

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- d. All equipment and materials used shall be fire retardant and shall comply with the requirements of the Local Fire Services Department.
- e. Pipework insulation shall generally comply with BS 5422 and BS 5970:1981.
- f. Preformed sectional insulation shall be secured to pipework by means of adhesive or approved equal providing 100% coverage on the internal face and corrosion resistant metal bands or fibreglass tape not less than 20mm wide spaced at 450 mm centre, with additional bands at all points where material has been shaped. All joints in the aluminium foil shall be sealed with a 75 mm wide pressure aluminium tape to provide a continuous external vapour seal. Under no circumstances may corrosive adhesives be used or flat sheets be modified to form the equal to moulded sections.
- g. All insulation finished and coverings, adhesives, mastics, coatings, sealers and primers shall be Classified as Class I surface spread of flame to BS 476 Part 7.
- h. All pipes located in plant rooms or that run exposed to view shall be enclosed in factory prefabricated galvanised sheet casings in addition to the vapour barrier. The casing shall be not less than 0.8mm thick for pipework of 150mm and above measured over the insulation and not less than 0.6mm thick on smaller pipework.  

Noted

All pipes exposed to weather shall be encased in 25mm cement plaster with weatherproof paint or alternatively galvanised jacketing as specified above if insulated.
- i. Valves, strainers, fittings, etc shall be insulated with site injected polyurethane with same thickness as pipe insulation. All insulated valves, strainers, flanges, fittings, etc. shall be provided with 0.6mm thick galvanised split boxes arranged for easy removal. The box is to enclose the valve handle and to have a lid for valve access. A canvas insertion shall be provided to allow the insulated boxes to be removed without damage done to the insulation. The insulation on the pipe immediately adjacent to flanges shall be neatly swaged off to allow for easy removal of bolts.
- j. At the point of support, specially prepared blocks of high density polyurethane foam material shall be positioned to ensure the integrity of the vapour barrier by bonding the supports to the insulation.
- k. Flexible connector shall insulated with 50mm thick fibre glass with double sided aluminium foil vapour barrier.

